

7. (a) A student is asked to prepare a standard solution of sodium carbonate of concentration $0.0500 \text{ mol dm}^{-3}$.

The first step involved in the preparation of a standard solution is to weigh the appropriate mass of solid in a weighing bottle.

Calculate the mass of sodium carbonate required and describe the remaining steps that the student should take to prepare 250 cm^3 of this standard solution. [6]

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- (b) The standard solution prepared in part (a) was used to determine the concentration of dilute hydrochloric acid.

25.0 cm³ samples of the sodium carbonate solution were titrated against the hydrochloric acid. These were the results.

Titration	1	2	3	4
Final reading / cm ³	26.50	26.80	26.20	26.55
Initial reading / cm ³	0.40	0.15	0.00	0.25
Titre / cm ³	26.10	26.65	26.20	26.30

- (i) Calculate the mean titre that should be used to determine the concentration of the hydrochloric acid. [1]

Mean titre = cm³

- (ii) The burette used in the titrations has an uncertainty for each reading of $\pm 0.05 \text{ cm}^3$. Estimate the maximum percentage error in the **titre** in titration 4. Show your working. [1]

Percentage error = %

- (iii) Apart from errors in reading the burette, suggest **one** reason why incorrect titres may have been obtained when carrying out the titrations. Explain the effect of this error on the value of the titre obtained. [2]

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- (c) A student wanted to verify that the ammonium sulfate content in a fertiliser was as given on the packet.

He boiled 4.24 g of the fertiliser with 50.0 cm³ of 1.00 mol dm⁻³ sodium hydroxide solution until no more ammonia was evolved.

The residual solution containing excess sodium hydroxide was made up to 250 cm³ with distilled water in a standard volumetric flask. A 25.0 cm³ portion of this solution required 22.65 cm³ of 0.090 mol dm⁻³ hydrochloric acid for neutralisation.

- (i) Calculate the number of moles of sodium hydroxide in the 250 cm³ flask and hence the number of moles that reacted with the fertiliser. [3]

$n(\text{NaOH}) = \dots\dots\dots \text{mol}$

- (ii) The equation for the reaction between ammonium sulfate and sodium hydroxide is:



Use your answer to part (i) to calculate the percentage by mass of ammonium sulfate in the fertiliser. [2]

Percentage = $\dots\dots\dots\%$



SECTION B

Answer all questions in the spaces provided.

7. (a) Melting temperatures vary down groups and across periods.

(i) Explain why chlorine is a gas but iodine is a solid at room temperature and pressure. [3]

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(ii) Explain why sodium has a lower melting temperature than aluminium. [1]

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(iii) Explain why silicon has a higher melting temperature than phosphorus. [1]

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(b) State and explain how you would expect the first ionisation energy of nitrogen to compare with the first ionisation energy of oxygen. [2]

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SECTION BAnswer **all** questions.

7. (a) The boiling temperatures of the hydrides of some Group 4 and Group 7 elements are shown in the table below.

Group 4 hydride	Boiling temperature / °C	Group 7 hydride	Boiling temperature / °C
CH ₄	–161	HF	20
SiH ₄	–112	HCl	–85
GeH ₄	–88	HBr	–66

Hydrogen has an electronegativity value of 2.1.

The electronegativity values of the Group 4 and Group 7 elements are given below.

Element	Electronegativity	Element	Electronegativity
C	2.5	F	4.0
Si	1.8	Cl	3.0
Ge	1.8	Br	2.8



For the hydrides of the Group 4 and Group 7 elements shown in the table opposite

- describe any trends and anomalies in boiling temperature
- explain any differences in boiling temperature in terms of the intermolecular forces present.

[6 QER]

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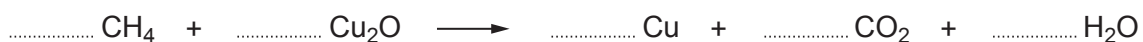
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(b) Methane reacts with copper(I) oxide according to the following equation.



(i) Balance the equation. [1]

(ii) Explain why this reaction is described as a redox process. [1]

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(c) Compounds containing a Group 1 metal, a Group 3 metal and hydrogen only are known as complex metal hydrides. There has been much interest in their use as hydrogen storage systems for future fuel cell-powered vehicles.

One such hydride contains 38.7% Li and 50.1% Al.

Find the empirical formula of this compound. [2]

Empirical formula

